

Goal

Develop a Bank Account System using OOP concepts in Python.

Approach

Step 1:

Created two custom exceptions:

- InsufficientFundsError
- OverdraftLimitError

Step 2:

Created a base class called BankAccount containing:

- Account owner
- Account balance
- Transaction history

Methods implemented:

- deposit()
- withdraw()
- get_balance()
- get_statement()
- transfer()

Step 3:

Made balance, owner, and transaction list private using double underscores (__). Prevents direct access from outside the class (encapsulation).

Step 4;

Inherited SavingsAccount from BankAccount and added the following:

- interest_rate attribute
- add_interest() method
- withdrawal allowed only if a minimum balance of 500 remains in the account

Step 5:

Inherited CurrentAccount from BankAccount and added;

- overdraft_limit attribute
- withdrawal allowed even if the balance becomes negative but only up to the overdraft limit

Step 6

Overrode the withdraw() method in both subclasses.

- SavingsAccount checks minimum balance.

Step 7:

Created a transfer() method to transfer money between accounts

1. withdraws money from the source account
2. deposits money into the target account